

Sparse Congruence-Preserving Perturbations of Dyadic Series

What gap statistics can tell us about Erdős Problem #251

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ABSTRACT

For each convergent dyadic series with nonnegative integer coefficients, we construct non-negative integer perturbations whose sums fill an interval and whose supports lie in one set of upper Banach density zero. The corrections satisfy any prescribed bound tending to infinity for all sufficiently large indices; for each fixed modulus, coefficient and partial-sum residues are eventually preserved. The construction varies adjacent corrections while keeping their ordinary sum fixed. In the prime-gap application with eventual correction bound $\log(n+3)$, it gives vanishing total variation error for blocks of length $o(\log \log X)$ sampled on $[X, 2X)$ and preserves the cumulative asymptotic $n \log n$. The cumulative positions need not be prime, and the irrationality question remains open.

1 Introduction

Let $p_0 = 2, p_1 = 3, \dots$ be the primes and put $g_n = p_{n+1} - p_n$. Erdős asked whether

$$\Pi = \sum_{n \geq 0} \frac{p_n}{2^{n+1}}$$

is irrational [1, p. 94][2, p. 62] [3, p. 103]. Summation by parts gives $\Pi = 2 + \sum_{n \geq 0} g_n 2^{-(n+1)}$, with convergence justified in Section 4. We study which properties of the gaps survive when the latter sum is changed to a prescribed value.

We prove a perturbation theorem for arbitrary convergent dyadic series with non-negative integer coefficients. All corrections are supported on one sparse set, chosen before the value of the sum. Their sizes may be eventually bounded by any prescribed function tending to infinity, while the sums still fill an interval. We also prescribe, before choosing the target, a cutoff for each modulus beyond which both the corrections and their partial sums are divisible by that modulus. For prime gaps the same construction retains the distributions of blocks of length $o(\log \log X)$ sampled in $[X, 2X)$ and gives cumulative positions asymptotic to $n \log n$. The irrationality of the actual prime series remains a distinct question: these comparison sequences need not have prime cumulative positions.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

We use a standard interval-covering argument for series with finite choices. Fridy's generalised-base lemma [5, p. 194] treats prescribed digit bounds and nonincreasing weights. Crmarić and Kovač [17, Lemma 4] give the finite-choice form used here, and Kovač and Tao [6, Lemma 5.1] use analogous intervals of reciprocal choices. We must arrange these choices on a sparse set while preserving cumulative congruences. To do so, we redistribute a fixed correction between two adjacent indices. The simplest useful calculation is

$$\begin{array}{c|cccc} (e_n, e_{n+1}) & (0, 6) & (2, 4) & (4, 2) & (6, 0) \\ e_n + e_{n+1} & 6 & 6 & 6 & 6 \\ 2^{n+2}(e_n 2^{-n-1} + e_{n+1} 2^{-n-2}) & 6 & 8 & 10 & 12. \end{array}$$

Moving 2 from the later index to the earlier one changes the weighted sum while leaving the ordinary sum fixed. If the cumulative correction before the pair is odd, an additional 1 at $n - 1$ clears its residue modulo 2. Since the four pairs have the same sum, the choice of pair leaves every later cumulative residue unchanged. We repeat this operation with nested moduli and increasingly separated pairs. The larger separations make the weighted contributions smaller, so we enlarge the range of each pair until the later contributions fill the gaps between consecutive present choices.

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, intervals of indices contain integers, and $\log$ is natural unless a base is displayed. A set $S \subseteq \mathbb{N}$ has *upper Banach density* zero if

$$\lim_{H \rightarrow \infty} \sup_{u \in \mathbb{N}} \frac{|S \cap [u, u + H)|}{H} = 0.$$

For integers $X, m \geq 1$, let $\mu_{a,X,m}$ denote the distribution of $(a_n, \dots, a_{n+m-1})$ when n is uniform on $[X, 2X)$. Blocks are unnormalised and counted with multiplicity. We use $d_{\text{TV}}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$.

Proposition 1.1 (sparse changes preserving congruences). *Let $a : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $A = \sum_{n \geq 0} a_n 2^{-(n+1)} < \infty$, let $K \in \mathbb{N}$, and let $f : \mathbb{N} \rightarrow \mathbb{R}$ tend to $+\infty$. There exist a set $S \subseteq [K, \infty)$ of upper Banach density zero and a nondegenerate interval $I \subset (A, \infty)$ such that, for every $r \in I$, there is an integer correction $e : \mathbb{N} \rightarrow \mathbb{N}$ satisfying*

$$\text{supp } e \subseteq S, \quad e_n \leq f(n) \text{ eventually}, \quad \sum_{n \geq 0} (a_n + e_n) 2^{-(n+1)} = r.$$

For each integer $q \geq 1$ there is a cutoff N_q , independent of r , such that $q \mid e_n$ and $q \mid \sum_{i < n} e_i$ for all $n \geq N_q$. For every $\varepsilon > 0$, the choice $f(n) = (\log(n + 3))^\varepsilon$ can be made with

$$|S \cap [X, 2X)| = O_\varepsilon \left(\frac{X}{\log \log X} \right), \quad d_{\text{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) \leq \frac{m|S \cap [X, 2X + m)|}{X}.$$

In particular the distance tends to zero for every integer-valued $m = m(X) \geq 1$ with $m(X) = o(\log \log X)$, uniformly over target values.

The set S , the interval I and every congruence cutoff are fixed before r is chosen. The correction e depends on r ; its support may occupy only part of S . Polynomially

growing nonnegative integer sequences satisfy the convergence hypothesis, whereas $a_n = 2^n$ does not. The prescribed bound may grow as slowly as $\log \log(n + 3)$. A bounded correction would have to vanish identically: a modulus exceeding the bound makes e eventually zero, and its constant cumulative sum must then be divisible by every positive integer.

Section 2 gives the construction, and Section 3 applies it to prime gaps. We then identify the actual tails and characterise their rationality by integral shifts. This leads to finite separation tests and to a sufficient condition involving two consecutive small differences. The [companion's literature discussion](#) contains the further comparisons, including finite subsums and other dyadic coefficient sequences.

2 Proof of Proposition 1.1

Proof. Fixed totals. At stage j we put a correction at $n_j - 1$ and a variable pair at $n_j, n_j + 1$. We keep the pair's ordinary sum fixed. Consequently, the residue to be cancelled at a later stage is independent of all earlier pair choices. Let $n_{-1} \geq K$, write $s_j = n_j - n_{j-1} \geq 4$, and take positive integers M_j with $M_j \mid M_{j+1}$. For now let D_j be a positive integer bounding the digit at stage j . We first describe the corrections for these parameters, then choose them to obtain both sparsity and an interval of sums.

Starting from $C_0 = 0$, define

$$c_j = (-C_j) \bmod M_j \quad (0 \leq c_j < M_j), \quad C_{j+1} = C_j + c_j + M_j D_j.$$

For integers $0 \leq d_j \leq D_j$, set

$$e_{n_j-1} = c_j, \quad e_{n_j} = M_j d_j, \quad e_{n_j+1} = M_j(D_j - d_j), \quad (1)$$

and put $e_n = 0$ elsewhere. The two sums of a triple are

$$\begin{aligned} \sum_{r=n_j-1}^{n_j+1} e_r &= c_j + M_j D_j, \\ \sum_{r=n_j-1}^{n_j+1} e_r 2^{-r-1} &= c_j 2^{-n_j} + M_j D_j 2^{-n_j-2} + d_j M_j 2^{-n_j-2}. \end{aligned}$$

The first identity verifies that C_j is the cumulative correction before $n_j - 1$, whatever the digits are. Thus all C_j and c_j can be computed before choosing a single d_j . The second identity shows that increasing d_j by one transfers M_j from $n_j + 1$ to n_j and increases the weighted sum by

$$w_j = M_j 2^{-n_j-2}.$$

After c_j has been inserted, the cumulative correction is divisible by M_j ; the two entries of the pair preserve this divisibility. Fix $q \mid M_j$. At index n_j both the coefficient correction and the cumulative correction are divisible by q . At every later triple, q divides C_j and M_j , so it also divides c_j and both pair entries. The intervening corrections are zero. Hence

$$q \mid e_n, \quad q \mid \sum_{i < n} e_i \quad (n \geq n_j).$$

The entry c_j need not be divisible by q , which is why it is placed at $n_j - 1$, before the cutoff. The index n_j depends on q and the chosen moduli, but not on the digits or the value to be attained.

Spacing and size. The variable part at stage j takes the values $0, w_j, \dots, D_j w_j$. To fill the gaps between these choices, we need the later variable parts to have total range at least w_j . This suggests making each later range a difference of successive dyadic scales. We therefore choose $D_i = 2^{s_i} - 1$, for which

$$D_i w_i = (2^{n_i - n_{i-1}} - 1) M_i 2^{-n_i - 2} = M_i (2^{-n_{i-1} - 2} - 2^{-n_i - 2}).$$

Because $M_i \geq M_j$ for $i > j$, replacing every later modulus by M_j gives the finite estimate

$$\begin{aligned} \sum_{i=j+1}^J D_i w_i &\geq M_j \sum_{i=j+1}^J (2^{-n_{i-1} - 2} - 2^{-n_i - 2}) \\ &= w_j - M_j 2^{-n_J - 2} \quad (J > j). \end{aligned} \tag{2}$$

The digit range $D_j = 2^{s_j} - 1$ therefore gives the required overlap estimate even when the separations increase. Every entry of (1) is at most $M_j 2^{s_j}$, so we must keep this quantity below the prescribed bound at the three indices. We must also have $s_j \rightarrow \infty$ and ensure that every fixed positive integer divides all sufficiently late M_j .

To allow a nonmonotone bound f , put

$$h(n) = \inf_{m \geq n} \min(f(m), m).$$

This nondecreasing lower envelope tends to infinity. A bound chosen using $h(n_{j-1})$ is therefore valid at each of the later coordinates of the triple. The extra cap by m serves a separate purpose: $e_n \leq n$ will make the weighted corrections summable. Choose n_{-1} with $h(n_{-1}) \geq 32$, and set

$$k_j = \max\{k \geq 2 : k! 2^{k+2} \leq h(n_{j-1})\}, \quad M_j = k_j!, \quad s_j = k_j + 2, \quad n_j = n_{j-1} + s_j.$$

The maximum exists: $k = 2$ is admissible because $2! 2^4 = 32$, and $k! 2^{k+2}$ tends to infinity. Since h is nondecreasing, so is k_j . Moreover $n_j \rightarrow \infty$, hence $h(n_j) \rightarrow \infty$ and $k_j \rightarrow \infty$. It follows that $M_j \mid M_{j+1}$, every fixed q divides all sufficiently late $M_j = k_j!$, and the separations $s_j = k_j + 2$ tend to infinity. The same choice gives the size estimate: at each supported coordinate n of the j th triple,

$$0 \leq e_n \leq M_j 2^{s_j} \leq h(n_{j-1}) \leq \min(f(n), n).$$

This proves the size bound and convergence for every choice of digits.

Let $S = \bigcup_{j \geq 0} \{n_j - 1, n_j, n_j + 1\}$. For any R , all but finitely many successive centres are at least R apart. An interval of length H therefore contains at most $3(H/R + 2)$ points of S , in addition to a fixed finite set. The bound is uniform in the position of the interval. Divide by H , take the supremum over positions and let $H \rightarrow \infty$; then let $R \rightarrow \infty$. This proves upper Banach density zero.

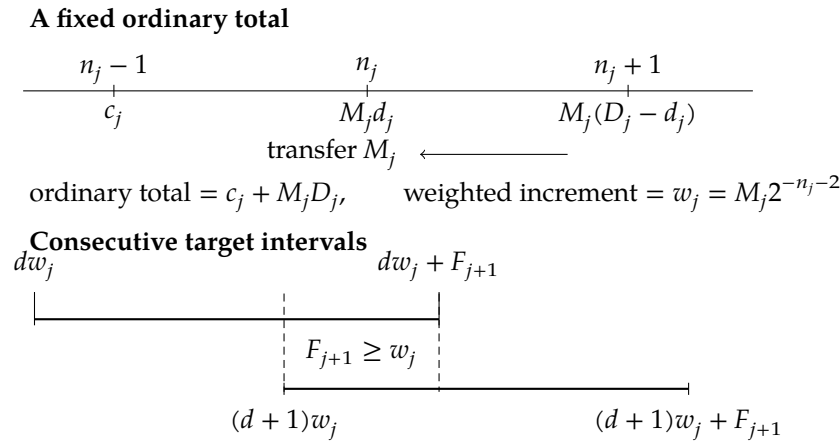


Figure 1: Redistributing M_j between adjacent entries keeps their ordinary sum fixed and increases the dyadic sum by w_j . The candidate target intervals in the lower panel cover without gaps when $F_{j+1} \geq w_j$. The sketch shows strict overlap; the proof also allows equality and establishes attainment by keeping the remainder in $[0, F_j]$ as $F_j \rightarrow 0$.

Attaining the interval. We have now fixed the centres, moduli, digit ranges and residue corrections, but have left every digit free. Define

$$\beta = \sum_{j \geq 0} (c_j 2^{-n_j} + D_j w_j), \quad F_j = \sum_{i \geq j} D_i w_i.$$

The first quantity is the contribution with all digits zero. The second is the maximum possible variable contribution from stage j onwards. Both are finite by the bounds just proved, with $\beta > 0$, $F_0 > 0$ and $F_j \rightarrow 0$. The total correction is

$$\sum_n e_n 2^{-n-1} = \beta + \sum_j d_j w_j.$$

Taking $J \rightarrow \infty$ in (2) now gives

$$F_{j+1} \geq M_j 2^{-n_j-2} = w_j. \quad (3)$$

Every variable sum from stage j onwards lies in $[0, F_j]$; we must still show that every point of this interval is attained. A target in $[0, F_j]$ leaves a remainder in $[0, F_{j+1}]$ after choosing digit d exactly when it belongs to

$$[dw_j, dw_j + F_{j+1}], \quad d = 0, \dots, D_j.$$

These intervals meet or overlap by (3), and their union is $[0, F_j]$ because $F_j = D_j w_j + F_{j+1}$. Figure 1 shows the two calculations. Given $x \in [0, F_0]$, start with $\rho_0 = x$. Having chosen the first j digits, choose d_j so that

$$\rho_{j+1} = \rho_j - d_j w_j \in [0, F_{j+1}].$$

The covering just proved makes this possible at every stage. Since $0 \leq \rho_j \leq F_j$ and $F_j \rightarrow 0$, we obtain

$$x - \sum_{i=0}^j d_i w_i = \rho_{j+1} \rightarrow 0.$$

Thus every $x \in [0, F_0]$ is attained, and we may take $I = (A + \beta, A + \beta + F_0)$. The target $r \in I$ determines only the digits, through $x = r - A - \beta$; it changes neither S , I nor the cutoff $N_q = n_j$ chosen using $q \mid M_j$. This last selection is the standard finite-choice covering argument of [17, Lemma 4]. The construction above supplies its summability and overlap hypotheses while imposing the congruences and requiring the support to lie in S .

The logarithmic scale. For $f(n) = (\log(n+3))^\varepsilon$, we can choose the separations explicitly at the scale $\log \log n$. Keep the same triples and $D_j = 2^{s_j} - 1$, and now choose

$$s_j = \left\lfloor \frac{\varepsilon}{2} \log_2 \log(n_{j-1} + 3) \right\rfloor,$$

$$k_j = \max\{k \geq 2 : k! \leq (\log(n_{j-1} + 3))^{\varepsilon/4}\},$$

with $M_j = k_j!$ and $n_j = n_{j-1} + s_j$. Start far enough out that $s_j \geq 4$ and all choices exist. The exponents $\varepsilon/4$ and $\varepsilon/2$ add to $3\varepsilon/4 < \varepsilon$, so the corrections satisfy the prescribed bound:

$$M_j 2^{s_j} \leq (\log(n_{j-1} + 3))^{3\varepsilon/4} \leq (\log(n+3))^\varepsilon$$

at each coordinate n of the triple. After enlarging the initial index, this is also at most n . The moduli remain nested, and every fixed positive integer divides all sufficiently late moduli. Summability, the congruence induction and (2) therefore apply without change. Here $s_j \asymp_\varepsilon \log \log n_j$, so

$$|S \cap [X, 2X)| = O_\varepsilon(X / \log \log X).$$

Choose the same uniform starting index $n \in [X, 2X)$ for the two blocks. A changed coordinate t can affect only the starts $n \in [t - m + 1, t]$, of which there are at most m . Only $t \in S \cap [X, 2X + m)$ can occur in a sampled block. The union bound consequently gives

$$d_{\text{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) \leq \mathbb{P}(e_{n+i} \neq 0 \text{ for some } 0 \leq i < m)$$

$$\leq \frac{m|S \cap [X, 2X + m)|}{X}.$$

For $m \leq X$, two dyadic bands cover $[X, 2X + m)$ and give the bound $O_\varepsilon(m / \log \log X)$. In particular it tends to zero when $m = o(\log \log X)$, uniformly over the values attained, since the same set S contains every correction support. The block length enters solely through the number of starts that one altered coordinate can affect.

□

The coupling also applies to tests depending on the starting index. If their absolute value is bounded by B_0 , their mean changes by at most $2B_0 m |S \cap [X, 2X + m)| / X$. For a function of the block alone, the bound is $2B_0 d_{\text{TV}}$. These are absolute errors: they give

no relative estimate for an event whose probability tends to zero, nor a comparison of the complete infinite tails. The companion's [Section 2](#) also distinguishes upper Banach density from ordinary density.

3 Application to prime gaps

The elementary bound $p_n \leq 1250(n+1)^4$ suffices for convergence of the prime and gap series. We use the central-binomial argument of Erdős [19, pp. 194–196]; [Appendix A of the companion](#) derives this bound, including the prime-power estimate. This supplies the convergence hypothesis needed to apply [Proposition 1.1](#) to the prime gaps.

Corollary 3.1 (a rational sum with the stated prime-gap statistics). *Let $p_0 = 2, p_1 = 3, \dots$ be the primes and $g_n = p_{n+1} - p_n$. Given $K \in \mathbb{N}$ and $0 < \varepsilon \leq 1$, there is $b : \mathbb{N} \rightarrow \mathbb{N}$ with rational dyadic sum such that $b_n = g_n$ for $n < K$, $b_n \geq g_n$, and $b_n - g_n \leq (\log(n+3))^\varepsilon$ eventually. For every fixed positive modulus, both the coefficients and the cumulative positions eventually retain their corresponding residues. The empirical distributions of unnormalised blocks have total variation distance tending to zero for lengths $o(\log \log X)$, and for every fixed nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$,*

$$|\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| = o(N).$$

Moreover, with $P_n = 2 + \sum_{i < n} b_i$,

$$0 \leq P_n - p_n = O_\varepsilon\left(\frac{n(\log(n+3))^\varepsilon}{\log \log n}\right), \quad P_n \sim n \log n.$$

These positions are not asserted to be prime.

Lean proves this for every $\varepsilon > 0$, assuming Schlage-Puchta's Lemma 4 [18] only for the nonconcentration bound and the prime number theorem only for $P_n \sim n \log n$.

Proof. Since the gap series converges, we can apply [Proposition 1.1](#) to $a = g$. Choose a rational r in its nondegenerate interval I and put $b = g + e$. The proposition gives all the correction and congruence assertions, together with the comparison of growing blocks. Fix k and a nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$. A new zero can occur only in a block meeting S , so

$$\begin{aligned} & |\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| \\ & \leq |\{n < N : F(g_n, \dots, g_{n+k}) = 0\}| + \sum_{i=0}^k |S \cap [i, N+i)|. \end{aligned}$$

The first count is $o(N)$ by Schlage-Puchta's Lemma 4 [18]; the finitely many support counts are $o(N)$ because S has density zero. Only changed blocks are counted, so the values of F need not be bounded. Since b was chosen before F, k , the same sequence satisfies every fixed-polynomial condition.

To estimate $P_n - p_n = \sum_{i < n} e_i$, first observe that $|S \cap [0, n)| = O_\varepsilon(n / \log \log n)$. Indeed, the indices below $\sqrt{n}$ contribute at most $\sqrt{n}$ points, and on $[\sqrt{n}, n)$ we sum the dyadic support bounds, whose lengths add to $O(n)$ and whose $\log \log X$ are comparable to $\log \log n$. Multiplying by the pointwise correction bound gives the stated error. For $0 <$

$\varepsilon \leq 1$ this is $o(n \log n)$, so the prime number theorem [12, Chapter 6] gives $P_n \sim n \log n$. Finally, summing nonnegative terms in the opposite order, we obtain

$$\sum_{n \geq 0} \frac{P_n}{2^{n+1}} = 2 + \sum_{i \geq 0} b_i \sum_{n > i} 2^{-n-1} = 2 + \sum_{i \geq 0} \frac{b_i}{2^{i+1}} = 2 + r \in \mathbb{Q}.$$

□

For $\varepsilon > 1$ we may use exponent 1, which gives smaller corrections and the error $O(n \log(n+3)/\log \log n)$. Thus the conclusions extend to every $\varepsilon > 0$. For each fixed $y \geq 2$, eventual congruence modulo $y!$ also gives $\gcd(P_n, y!) = 1$ once $p_n > y$. The cutoff depends on y , so this excludes no growing range of possible prime divisors. Even a prime-valued sequence of positions could omit primes.

Land's draft [8, Theorem 2] proves conditional irrationality under a uniform Hardy–Littlewood hypothesis of the kind formulated by Kuperberg [7, Conjecture 1.3]. Ringer's draft [16] obtains conditional normality. Their quantitative prime-pattern hypotheses are stronger than the absolute block comparison above and are not assumed here. The companion's Section 3 compares the hypotheses and sampling conventions.

4 The prime series and its actual tails

We return to the actual primes and write $G = \sum_{n \geq 0} g_n 2^{-(n+1)}$. To use a recurrence for these sums, we must retain the endpoint in finite summation by parts and then show that it vanishes. The polynomial bound from Section 3 supplies this limit and convergence of both series.

Theorem 4.1 (prime-to-gap identity). *The actual prime and gap series satisfy $\Pi = 2 + G$. Their complete tails*

$$T_N = \sum_{j \geq 1} g_{N+j} 2^{-j}$$

satisfy $T_0 = 2G - 1$ and $T_{N+1} = 2T_N - g_{N+1}$. Thus Π , G and T_0 have the same rationality status.

Proof. Finite summation by parts gives

$$\sum_{i=0}^n \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

Since $p_n = O((n+1)^4)$, the endpoint tends to zero and we obtain $\Pi = 2 + G$. We obtain the tail recurrence by splitting off the first term of T_N ; removing $g_0 = 1$ from G gives $T_0 = 2G - 1$. □

Tao noted this reduction in the problem's forum discussion [15, 7 October 2025].

Corollary 4.2 (exact irrationality reformulation). *The prime-value dyadic series is irrational if and only if the prime-gap dyadic series is. Both series converge by the polynomial bound proved above; neither side is proved irrational.*

Proof. Adding the rational number 2 preserves rationality and irrationality. $\square$

The recurrence alone does not identify the infinite tails: adding $C2^N$ to any solution gives another solution. We distinguish the actual tails by the vanishing condition in the next lemma.

Lemma 4.3 (the boundary condition for the tail recurrence). *Let $\sum_{j \geq 1} |a_j| 2^{-j} < \infty$ and $U_{N+1} = 2U_N - a_{N+1}$. Then $U_N = \sum_{j \geq 1} a_{N+j} 2^{-j}$ for every N if and only if $2^{-N} U_N \rightarrow 0$.*

Proof. Iteration gives $2^{-N} U_N = U_0 - \sum_{j=1}^N a_j 2^{-j}$. The limit is zero exactly when U_0 is the full series. Subtracting its first N terms and multiplying by 2^N then gives the tail formula. Conversely the formula implies $2^{-N} U_N = \sum_{k > N} a_k 2^{-k} \rightarrow 0$ by absolute convergence. $\square$

4.1 Integral shifts

For the recurrence $U_{N+1} = 2U_N - a_{N+1}$ with $a_n \in \mathbb{Z}$, write $D_h(N) = U_{N+h} - U_N$. Modulo integers, each step is doubling: U_N has the fractional part of $2^N U_0$. An integral shift means that a fractional part has returned to an earlier value. Such a return makes $2^N(2^h - 1)U_0$ an integer and forces rationality when $h > 0$. Conversely, doubling a rational number eventually gives periodic fractional parts. The next theorem records the denominator calculation and the exact quantifiers.

Theorem 4.4 (exact rationality classification). *For a real integer-coefficient recurrence, the following are equivalent: $U_0 \in \mathbb{Q}$; $D_h(N) \in \mathbb{Z}$ for some $h \geq 1, N \geq 0$; and, for some fixed $h \geq 1$, $D_h(N) \in \mathbb{Z}$ at every sufficiently large N . More precisely, if $U_0 = u/(2^s d)$ is in lowest terms, with d odd, then*

$$\text{den}(U_N) = 2^{\max(s-N, 0)} d, \quad D_h(N) \in \mathbb{Z} \iff N \geq s \text{ and } d \mid 2^h - 1 \quad (h \geq 1).$$

Consequently U_0 is irrational exactly when every positive shift is nonintegral at every index, equivalently when for each fixed $h \geq 1$ there are arbitrarily late nonintegral shifts.

Proof. Iteration gives

$$U_N - 2^N U_0 \in \mathbb{Z}, \quad D_h(N) - 2^N(2^h - 1)U_0 \in \mathbb{Z}.$$

An integral shift with $h \geq 1$ makes the nonzero integer multiple $2^N(2^h - 1)U_0$ integral, and therefore forces $U_0 \in \mathbb{Q}$. Conversely, write $U_0 = u/(2^s d)$ in lowest terms. Multiplication by 2^N cancels exactly $\min(N, s)$ powers of two from the denominator. The remaining factor $2^h - 1$ is odd, so the second relation is integral exactly when $N \geq s$ and $d \mid 2^h - 1$. For $d > 1$, Euler's congruence provides such a shift $h = \varphi(d)$, and for $d = 1$ we take $h = 1$. This shift works for every $N \geq s$. Negating the pointwise and eventual criteria gives the two assertions about irrationality. $\square$

The theorem applies to the recurrence itself, without the boundary condition of Lemma 4.3 or any assumption of primality. The eventual periodicity is that of fractional parts, and need not hold for the full tails. For instance, the positive even coefficients 2, 4, 2, 4, ... starting at $n = 1$ give tails 8/3, 10/3 alternately. Every shift of length 1 is nonintegral and every shift of length 2 is zero. This explains why a single shift length is insufficient.

5 Small tail differences and finite tests

Fix $h \geq 1$ and put $D_N = T_{N+h} - T_N$ and $\delta_N = g_{N+h+1} - g_{N+1}$. Then $D_{N+1} = 2D_N - \delta_N$, where δ_N is even. Two differences of absolute value less than one can both be integral only if they vanish, in which case $\delta_N = 0$. More precisely, we have the following result.

Proposition 5.1 (two consecutive differences of absolute value less than one). *Let $D, D' \in \mathbb{R}$, $\delta \in 2\mathbb{Z}$ and $D' = 2D - \delta$. The conditions $|D| < 1$, $|D'| < 1$ and $\delta \neq 0$ hold exactly when, for some $s \in \{-1, 1\}$,*

$$\delta = 2s, \quad \frac{1}{2} < sD < 1.$$

In that case $sD' \in (-1, 0)$ and both D and D' are nonintegral.

Proof. The bounds on D, D' imply $-3 < \delta < 3$, so a nonzero even δ equals $2s$ with $s \in \{-1, 1\}$. Substituting in $D' = 2D - 2s$ gives $1/2 < sD < 1$. Conversely this interval gives $sD' = 2sD - 2 \in (-1, 0)$. $\square$

For $\delta = 2$, the endpoints $D = 1/2, 1$ give $D' = -1, 0$ respectively. The zero triple $D = D' = \delta = 0$ shows why the mismatch is required.

Problem 5.2 (two small tail differences at arbitrarily large indices). For every integer $h \geq 1$ and every cutoff N_0 , exhibit $N \geq N_0$ with

$$|T_{N+h} - T_N| < 1, \quad |T_{N+h+1} - T_{N+1}| < 1, \quad g_{N+h+1} \neq g_{N+1}. \quad (4)$$

A solution would give a nonintegral shift arbitrarily late for every h , and hence irrationality by Theorem 4.4. All three conditions must hold at the same index; separate infinite sets of small-difference and mismatch witnesses do not suffice.

By Proposition 5.1, such an index must have $\delta_N = \pm 2$. Schlage-Puchta's lemma [18, Lemma 4], applied to $x_h - x_0 \pm 2$, shows that these eligible indices already have density zero. Thus the required witnesses concern a rare event whose relative frequency is not controlled by the absolute block comparison in Proposition 1.1.

5.1 Finite separation from the integers

To decide nonintegrality from finitely many gaps, we bound the part of the difference that has not been summed. Choose $M(n) \geq g_n$ with $\sum_n M(n)2^{-n} < \infty$, and write

$$S_{h,N,L} = \sum_{j=1}^L (g_{N+h+j} - g_{N+j})2^{-j}, \quad R_{h,N,L}(M) = \sum_{j>L} (M(N+h+j) + M(N+j))2^{-j}.$$

Splitting the complete tails after L terms gives

$$D_N = S_{h,N,L} + 2^{-L}D_{N+L}, \quad |D_N - S_{h,N,L}| \leq R_{h,N,L}(M).$$

The uncomputed part is another complete tail difference, with the same shift h . The criterion below uses only the displayed bound; a general real majorant need not make that bound computable.

For the actual prime gaps, one computable choice is $M(n) = 1250(n + 2)^4$. For $P(x) = x^4 + 8x^3 + 36x^2 + 104x + 150$, summing the identity $2P(x) = (x + 1)^4 + P(x + 1)$ with dyadic weights gives $\sum_{j \geq 1} (x + j)^4 2^{-j} = P(x)$ and hence

$$R_{h,N,L}(M) = 1250 2^{-L} (P(N + h + L + 2) + P(N + L + 2)).$$

Proposition 5.3 (finite separation criterion). *If for every $h \geq 1$ and every cutoff N_0 there are $N \geq N_0, L \geq 1$ with*

$$\text{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M), \quad (5)$$

then Π is irrational.

Proof. The enclosure gives $\text{dist}(D_N, \mathbb{Z}) \geq \text{dist}(S_{h,N,L}, \mathbb{Z}) - R_{h,N,L}(M) > 0$. Thus each fixed positive shift is nonintegral at arbitrarily large indices, and Theorem 4.4 applies. $\square$

Remark (what arbitrary truncation does not gain). Whenever $|D - S_L| \leq R_L \rightarrow 0$, nonintegrality of D is equivalent to $\text{dist}(S_L, \mathbb{Z}) > R_L$ for some L . Indeed, for $\delta = \text{dist}(D, \mathbb{Z}) > 0$, take $R_L < \delta/2$ and use the 1-Lipschitz property of distance. Unrestricted depth therefore gives an exact reformulation, not an additional analytic estimate. At a prescribed logarithmic depth one must also control the margin from the integers as N varies; a shrinking remainder alone does not do so. The companion, Appendix D.2, gives an explicit example distinguishing refinement at a fixed value from enclosures of changing values.

Separation from $\mathbb{Z}$ does not specify either interval in Proposition 5.1. To certify (4), suppose $\delta_N = 2s$, where $s \in \{-1, 1\}$, and $|QD_N - A| \leq B$, with $A \in \mathbb{Z}$, $Q > 0$, $B \geq 0$ integers. Then sD_N lies in $[(sA - B)/Q, (sA + B)/Q]$. This whole interval lies in $(1/2, 1)$ precisely when its lower endpoint exceeds $1/2$ and its upper endpoint is less than 1, that is,

$$2sA - Q > 2B, \quad Q - sA > B.$$

Reducing A modulo Q would preserve its distance to $Q\mathbb{Z}$ but lose the enclosure's location. We retain the signed numerator and use the sign s fixed by the gap difference. At $h = 1, N = 2, L = 40$, one has $\delta_2 = g_4 - g_3 = -2$ and

$$Q = 2^{40}, \quad A = -662838684750, \quad B = 11764181250, \quad s = -1.$$

Here $A/Q = S_{1,2,40}$ and $B/Q = R_{1,2,40}(M)$, and the two strict margins are 202637379224 and 424908761776. The [companion's integer certificates](#) give the prime-index calculation and remainder derivation. This finite pair leaves the quantifiers in Problem 5.2 unresolved.

6 Further consequences and questions

We can place the interval in Proposition 1.1 inside $(A, A + \eta)$ for any $\eta > 0$. Choose $K' \geq K$ such that $(K' + 1)2^{-K'} < \eta$, and start the construction after K' . The bound $e_n \leq n$ from the proof gives

$$0 < \sum_n e_n 2^{-(n+1)} \leq (K' + 1)2^{-K'} < \eta.$$

This bounds the dyadic change, without bounding the ordinary total.

6.1 Nonperiodic coefficients

The usual periodicity criterion for rational binary expansions assumes digits in $\{0, 1\}$. With unrestricted integer coefficients, the following telescoping identity already gives a different behaviour:

$$\sum_{i=0}^{n-1} \frac{i-1}{2^{i+1}} = -\frac{n}{2^n} \rightarrow 0.$$

Its coefficients are unbounded and its first coefficient is -1 . We obtain a positive even example by taking $c_n = 2(n^2 + 4n + 2)$ and $U_n = 2(n + 4)^2$. Direct substitution gives $U_{n+1} = 2U_n - c_{n+1}$, while the polynomial growth gives $2^{-n}U_n \rightarrow 0$. Lemma 4.3 therefore identifies U as the complete tail and yields

$$\sum_{j \geq 1} c_j 2^{-j} = 32, \quad \sum_{n \geq 0} c_n 2^{-(n+1)} = 18.$$

The coefficients increase strictly and $c_{n+1} - c_n = 4n + 10$ is never ± 2 . The companion's Section 8 gives further examples, including a bounded nonperiodic sequence in Appendix D. None of these examples constructs consecutive primes.

Proposition 6.1 (prime gaps do not become periodic). *For every positive h , the actual consecutive-prime-gap sequence is not eventually periodic with period h .*

Proof. If the gaps were eventually periodic, they would be bounded. For each $m \geq 2$, however, the $m-1$ integers $m! + 2, \dots, m! + m$ are composite. The consecutive primes on either side therefore have a gap of at least m , contradicting boundedness. $\square$

The results on bounded prime gaps [13], clusters of each fixed size [11, 20], and extreme large gaps [4] do not give the simultaneous inequalities in (4). This remains so with Polymath's bound $\liminf_n g_n \leq 246$ [20, Theorem 1.4(i)]. For fixed h and $\varepsilon > 0$, taking $L = \lceil (4 + \varepsilon) \log_2(N + 2) \rceil$ in the polynomial remainder gives an error $O_{h,\varepsilon}(N^{-\varepsilon})$. The finite sum must still be separated from the relevant boundary by more than this error. For the actual prime series, nonintegral shifts are needed arbitrarily late for every positive length. Problem 5.2 would supply them. The perturbation theorem concerns the insufficiency of the specified congruences and block statistics, and leaves possible arguments using primality or quantitative rare-event estimates untouched.

7 Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The linked Lean declarations use Lean 4 [9] and mathlib [10]. The formal proof of Proposition 1.1 changes one coordinate at each stage, whereas the proof here uses triples. Both choose the set containing the supports, the interval and the congruence cutoffs before the value of the sum. For Corollary 3.1,

the formal statement covers every $\varepsilon > 0$ and assumes Schlage-Puchta’s lemma only for nonconcentration and the prime number theorem only for the asymptotic $P_n \sim n \log n$. The source links record formal support for the linked statements, rather than verification of the different printed proofs. The companion’s Appendix G retains the complete source index at this note’s original pin, including the finite telescoping identities and their nonperiodicity consequences. Its Section 3 retains the broader subsum and automatic-sequence comparisons.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Proposition 1.1	Lean · Comparator	Corollary 3.1	Lean [†]
Theorem 4.1	Lean · Comparator	Corollary 4.2	Lean · Comparator
Lemma 4.3	Lean · Comparator	Theorem 4.4	Lean · Comparator
Proposition 5.1	Lean · Comparator	Proposition 5.3	Lean · Comparator
Proposition 6.1	Lean · Comparator		